

# Despotism, Replicators, and Social Design: Betzig's Evidence as Empirical Foundation of the Extended Phenotype Theory of Law

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## ABSTRACT

Evolutionary accounts of legal institutions have largely treated empirical evidence from political anthropology as background context rather than as a direct test of theoretical predictions. This paper integrates Laura L. Betzig's cross-cultural findings on despotism and differential reproduction (1986) with Extended Phenotype Theory (EPT) and multilevel Evolutionary Game Theory (EGT) to establish a two-level explanatory architecture for the origin and transformation of normative systems. Betzig's statistical demonstration that hierarchical power predicts biased conflict resolution, which in turn predicts polygyny rate ( $p < 0.001$ ,  $N = 186$  pre-industrial societies), provides proximate-level evidence for EPT's central claim: normative institutions are not products of rational social contract but extended phenotypes of competing replicators operating under selection. I argue that proto-normative systems functioned as friction-reduction mechanisms for intra-coalition conflict — not, as is commonly assumed, as mechanisms for eliminating reproductive competition, but as mechanisms for managing it in ways that preserved coalition integrity while simultaneously encoding differential reproductive advantage for dominant actors.

Four analytical contributions follow. First, the paper distinguishes ultimate-cause analysis (gene fitness, Betzig's framework) from proximate-cause analysis (memetic replicators, EPT), showing the two levels are structurally analogous but analytically non-reducible. Second, it formalizes the Gendarme-to-Welfare transition as a morphospace mutation that introduces constructivist design assumptions into a system that evolved under spontaneous order constraints, applying the Pentland Impossibility Theorem (Lerer, 2026) to explain systematic implementation failure. Third, it identifies the Dennett-Nash Gap (Lerer, 2026) as the micro-mechanism linking institutional design failure to the Parasitic Spontaneous Order (PSO) documented empirically in compliance behavior. Fourth, it shows that twenty-first century identity claims function as high-efficiency coalitional signaling strategies whose fitness derives precisely from semantic imprecision, a pattern explained by Lateral Silencing Exaptation (Lerer, 2026) and confirmed by computational corpus analysis across six institutional

environments. Three falsifiable predictions are derived and distinguished from prior EPT predictions.

*Keywords:* extended phenotype, despotism, differential reproduction, proto-normativity, evolutionary game theory, multilevel selection, compliance theater, identity politics, memetic fitness, coalitional signaling

*JEL Classification:* K00, D03, Z13, H10

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## **1. Introduction: The Empirical Problem Evolutionary Legal Theory Has Avoided**

There is an observable pattern in pre-industrial societies that comparative institutional analysis has consistently underweighted. When Laura Betzig published *Despotism and Differential Reproduction* in 1986, she documented something uncomfortable for contract-based theories of law: across 186 pre-industrial societies, the degree of hierarchical political power did not predict better dispute resolution for the population. It predicted better reproductive outcomes for the person doing the resolving. The correlation between despotic authority and polygyny rate was not incidental. It was the point.

Betzig's data showed that hierarchical power predicts biased conflict resolution, which predicts differential reproduction at rates statistically indistinguishable from zero probability under the null hypothesis. Heads of state in highly stratified societies held simultaneous access to hundreds or thousands of women; societies with no formal hierarchy showed no harems above seven (Betzig, 1986, p. 97). The implication, which Betzig drew explicitly, is that legal institutions in pre-industrial contexts did not emerge from a social contract to promote general welfare. They emerged because men with greater force could enforce their own interests (Betzig, 1986, p. 78), and those interests were, at the level of ultimate cause, reproductive.

Evolutionary theories of law have engaged this evidence selectively. The existing EPT corpus treats proto-normative systems primarily through the lens of coalition dynamics, Dunbar's number, and deontic language emergence (Lerer, 2025a, 2025b), without directly confronting what Betzig's data imply about whose fitness the proto-normative system was actually serving. This paper addresses that gap.

The argument proceeds in four stages. Section 2 develops the two-level explanatory architecture that makes EPT and Betzig's framework compatible without collapsing one into the other. Section 3 analyzes the gendarme-to-welfare transition as a morphospace mutation with predictable failure modes. Section 4 applies the Dennett-Nash Gap to the same transition.

Section 5 examines twenty-first century identity claims as the latest instantiation of a selection dynamic visible in Betzig's earliest cases.

## **2. Two Levels, One Structure: Betzig and EPT as Complementary Frameworks**

### **2.1 The Level Problem**

Any attempt to integrate Betzig's sociobiological framework with EPT faces an immediate methodological problem. Betzig operates at the level of genetic ultimate causes: the claim is that power, prestige, and legal authority are proxies for reproductive fitness, and that natural selection has produced organisms that seek them as means to differential reproduction. The unit of analysis is the individual and the gene. EPT, by contrast, operates at the level of cultural replicators: the claim is that legal institutions are extended phenotypes of memes competing for transmission, and that their persistence is explained by memetic fitness rather than genetic advantage or normative validity (Lerer, 2025a).

These are not competing explanations. They are explanations at different levels of the same phenomenon. Betzig identifies the ultimate selective pressure that shaped the psychological architecture underlying early normative behavior. EPT identifies the proximate replicator that encodes and transmits normative patterns across individuals and generations. The distinction maps directly onto Tinbergen's (1963) four questions: Betzig answers why (ultimate function) and how it evolved (phylogeny); EPT answers how it works now (mechanism) and how it develops (ontogeny at the cultural level).

This distinction matters because collapsing the two levels produces a theoretical error: it assigns teleological ultimate-cause reasoning to the memetic system. To say that "the objective of the normative system is to avoid sexual competition" is to mistake Betzig's ultimate-cause finding for EPT's proximate-cause mechanism. The meme does not have reproductive goals. It has transmission fitness. The gene has reproductive goals. The two can produce identical institutional structures through different causal pathways.

### **2.2 What Betzig's Data Actually Show for EPT**

Read as evidence for EPT rather than for sociobiology per se, Betzig's cross-cultural data provide three specific empirical constraints that prior EPT formulations have not addressed.

First, proto-legal institutions were not neutral arbiters. Betzig documents that as hierarchical complexity increased, arbiters accumulated perquisites in the form of fines, fees, bribes, and confiscations (Betzig, 1986, p. 55). In one Dahomean case she cites, a provincial head was described as "great at embezzlement, and woe betide the litigious wight whose case falls into his hands" (Betzig, 1986, p. 56, citing Herskovits). This is not exceptional; Betzig shows it is the statistical norm. For EPT, this means that the proto-legal extended phenotype was not, at

its origin, a neutral coordination technology. It was a meme complex whose transmission fitness was parasitically coupled to the fitness of its most powerful human vehicles.

Second, the normative regulation of sexual behavior — specifically, norms prohibiting adultery with the wives of high-status men, punishable by death, castration, or collective family execution — is documented across polities from Ashanti to Inca to Byzantine (Betzig, 1986, chapters 3–5). These norms were enforced with a consistency and severity that far exceeded enforcement of property norms or norms regulating violence between subordinates. For EPT, this pattern identifies sexual-access protection as one of the earliest and most stable extended phenotypic effects of power-aligned memplexes.

Third, Betzig demonstrates that manipulation of kinship classification and incest taboos followed reproductive logic rather than normative consistency. Men with social advantages broke prescriptive marriage rules to expand their pool of eligible wives; the Kapauku case she documents in detail shows a headman contracting incestuous unions that his own kin accepted because coercion made resistance unprofitable (Betzig, 1986, p. 74). For EPT, this establishes an early empirical instance of what the program has formalized as the Capacity-Gap ( $\Delta = \text{TCI} - \text{CLI}$ ): when the ultimate-cause pressure is sufficient, the meme encoding the normative constraint is simply displaced by a competing meme that better serves the dominant vehicle's fitness.

### **2.3 The Coalition Integrity Mechanism Reframed**

The standard EPT account of proto-normative systems emphasizes coalition integrity: proto-norms reduced intra-coalition friction, enabled cooperation beyond dyadic reputation tracking, and created selection pressure for deontic language as a scaling technology (Lerer, 2025b). This account is correct at the level of the subordinate majority in Betzig's data. For the non-elite members of a hierarchy, norms preventing unconstrained violence by the physically strongest members were genuine welfare improvements. The bonobo female coalitions documented by Surbeck et al. (2025) — females form aggressive coalitions targeting males in 85% of cases and, when doing so with greater frequency, outrank males in the dominance hierarchy — constitute the primate analogue cited in EPT's genealogy.

But Betzig's data require adding a second mechanism operating simultaneously. While the subordinate majority benefits from friction reduction, the dominant minority benefits from asymmetric enforcement. These are not contradictory functions; they are two fitness payoffs from the same institutional structure. The gendarme-state proto-norm package delivers coalition stability to the group and reproductive advantage to the dominant actor. This is why proto-normative systems could scale and persist: they produced genuine coordination benefits at the population level while simultaneously encoding elite reproductive interests as protected entitlements. The meme was fit on both levels simultaneously.

### **3. From Friction Reduction to Active Design: The Morphospace Mutation**

#### **3.1 What the Gendarme State Actually Was**

The phrase "gendarme state" in political theory refers to a minimal institutional configuration that enforces property rights and resolves disputes without attempting to engineer social outcomes. In the evolutionary vocabulary developed in this research program, the gendarme configuration is characterized by ex-post liability, negative selection only (removing harmful interactions rather than specifying positive outcomes), and enforcement through third-party punishment of norm violators rather than constructivist redistribution.

Betzig's evidence suggests that even this minimal configuration was never normatively neutral. The "gendarme" at the head of a pre-industrial hierarchy was, in her framework, simultaneously extracting reproductive resources and providing genuine coordination services. The equilibrium persisted because the coordination services were real enough that subordinates' net welfare under the system exceeded their welfare under constant uncoordinated conflict, even accounting for the extraction.

For EPT, this establishes the gendarme configuration as a Parasitic Spontaneous Order (PSO) in the specific sense defined in prior work (Lerer, 2026e): an institutional arrangement that produces genuine coordination benefits while simultaneously allowing organized extraction by actors who control enforcement. The extraction is not a bug of the gendarme system. It is the condition that made the gendarme system evolutionarily stable, because it gave the dominant actor strong selective incentives to maintain the system.

#### **3.2 The Constructivist Mutation**

The transition to active social design — what political theory calls the welfare state in its various instantiations — represents a mutation in the institutional morphospace rather than a development along a continuum. The conceptual shift is from ex-post coordination (enforcing what agents agreed to, resolving disputes that arise) to ex-ante outcome engineering (specifying what distribution of resources or opportunities is required, and enforcing it constructively).

This transition introduces a theoretical contradiction that Hayek identified at the epistemological level and that EPT formalizes as the Pentland Impossibility Theorem (Lerer, 2026c): under dispersed knowledge and heterogeneous preferences, no designed coordination mechanism can simultaneously satisfy participation constraints, incentive compatibility, and goal achievement without coercion. The welfare-state configuration attempts to solve a problem — resource distribution — using a tool — constructivist institutional design — that requires information the designer cannot possess.

Three empirical cases in the prior EPT corpus document the magnitude of this failure. UK Data Trusts achieved 0% operational deployment despite funding above £2 million. EU ETS Phase I experienced a 99.7% price collapse from over-allocation of permits. GDPR produced a 20% increase in data management costs with no measurable privacy improvement (Lerer, 2026c). Each represents a case where constructivist design produced the opposite of its intended effect, not because of corruption or implementation failure, but because of structural incompatibility between the design assumption (controllable system) and the actual dynamics (evolving memetic ecology).

Prior work in this program has operationalized this dimension through the Constitutional Lock-in Index (CLI), a measure of structural transmission rigidity calibrated across 193 jurisdictions (Lerer, 2026d); domains above the ESS threshold of 0.71 exhibit resistance to reform independent of political will.

### **3.3 Betzig's Contribution to This Transition**

Betzig's framework illuminates why the welfare-state transition is not simply an epistemological error. It is also a shift in which actors' reproductive and resource interests the institutional meme encodes. The gendarme system encoded elite reproductive interests while providing subordinate coalition benefits. The welfare-state claim, whatever its normative valence, is that the institutional meme should encode majority interests rather than elite interests.

This is, in evolutionary terms, a frequency-dependent strategy shift. The question EPT asks is not whether the shift is normatively desirable but whether the new meme complex has sufficient fitness to persist against the countervailing selection pressure from actors who benefited from the prior configuration. Betzig's data predict the answer: in any system where the actors administering and enforcing the new institutional rules have access to extraction opportunities, the PSO dynamic will re-emerge. The welfare-state bureaucracy becomes, under standard Betzig-EPT analysis, a new vehicle for differential resource accumulation by those controlling the enforcement apparatus, with the population-level coordination benefits again serving as the evolutionary subsidy that keeps the system stable enough to persist.

## **4. The Dennett-Nash Gap at the Welfare State Scale**

### **4.1 The Intentionality Mismatch in Modern Institutional Design**

EPT's Asymmetric Intentionality Theory (AIT, Lerer, 2025c) classifies agents by their capacity for recursive social reasoning. Level 3 agents form beliefs about others' beliefs about their own beliefs, enabling shame, reciprocity, moral reasoning, and responsiveness to reputational mechanisms that operate through mutual mental-state attribution. Level 1 agents optimize observable payoffs — including regulatory compliance scores — without this recursive architecture. Corporations, bureaucracies, and algorithms are designed Level 1 optimizers: they are constructed by Level 3 humans to behave as Level 1 maximizers.

Welfare-state regulation consistently embeds Level 3 assumptions about the agents it regulates. Integrity programs under Argentina's Ley 27.401, GDPR compliance regimes, ESG reporting requirements, and DEI implementation mandates all presuppose that regulated entities will internalize normative commitments through moral reasoning. A corporation that has genuinely internalized an anti-corruption commitment will behave differently from one that is producing compliance theater because the expected cost of getting caught exceeds the expected benefit of the violation. Regulatory regimes designed for the first type of actor fail systematically when the actual population contains a substantial fraction of the second type.

Betzig's contribution here is to show that this mismatch is not a recent phenomenon of modern regulatory complexity. The Dahomean provincial head who was "great at embezzlement" was already performing the earliest documented version of compliance theater: providing the coordination service (dispute resolution) required to maintain his position while simultaneously extracting the maximum available perquisites. The form has modernized; the mechanism is the same.

## **4.2 Compliance Theater as Level 1 Response to Level 3 Design**

The Generalized Intentionality Mismatch Theorem (Lerer, 2025c) predicts five failure modes when a legal regime designed for Level 3 actors is applied to a population containing a substantial fraction of Level 1 actors: compliance theater, letter-versus-spirit exploitation, recidivism, regulatory gaming, and social enforcement failure. Each failure mode has a precise Betzig-era analogue.

Compliance theater appears in Betzig's data as the elaboration of formal normative structures (kinship classification, marriage rules, incest taboos) that powerful actors manipulate while publicly maintaining. The Kapauku headman who contracted incestuous unions while remaining the society's normative authority was performing compliance theater in its earliest documented form: maintaining the formal institutional signal while optimizing the underlying behavior for his own fitness.

Letter-versus-spirit exploitation appears in the norm that adultery with a chief's wife is capital but adultery with a common wife is negotiable — a split that Betzig documents across multiple polities. The legal text prohibits adultery; the enforcement architecture distinguishes based on whose reproductive interests are at stake.

The meme encoding these asymmetries does not require anyone to consciously design them. Selection pressure maintains institutional configurations where enforcement asymmetry correlates with enforcement-agent fitness, because those configurations are evolutionarily

stable in Maynard Smith's sense: no actor with control over enforcement has an individual incentive to deviate.

## **5. Identity Claims as High-Fitness Coalitional Signals: A Betzig-EPT Synthesis**

### **5.1 The Continuity Problem**

Twenty-first century DEI claims and identity politics activism present a theoretical problem for purely gene-level analysis. Betzig's framework predicts that coalitional behavior is driven by reproductive fitness: individuals form coalitions to access resources and reproductive opportunities. Identity claims in contemporary democratic societies do not map straightforwardly onto individual reproductive strategy.

The resolution lies in the two-level architecture developed in Section 2. At the gene level, Betzig's predictions apply most directly to pre-industrial and early-state societies where the correlation between institutional power and reproductive access was direct and measurable. At the meme level, EPT's predictions apply to the current environment, where the relevant fitness currency is transmission success across minds and institutions rather than offspring count.

The question is not whether DEI meme proponents are optimizing for reproductive fitness in Betzig's sense. The question is whether the DEI meme complex has properties that give it high transmission fitness in contemporary institutional environments. The answer, supported by computational corpus analysis across six institutional environments in prior EPT work, is yes, and the mechanism is lexical exaptation rather than argumentative superiority (Lerer, 2026f).

### **5.2 Lexical Exaptation and the Abandoned Niche**

Betzig documents a pattern in political anthropology that is structurally identical to what the EPT corpus documents for normative concepts: powerful memes occupy niches that become available through the exit of prior occupants. The institutional niche for "resolution of inter-individual conflict" was occupied, in Betzig's pre-industrial polities, by whichever actor had the greatest enforcement capacity. When state structures developed, they displaced local arbiters not through superior normative legitimacy but through superior organizational capacity.

The lexical exaptation documented for "equity" follows the same logic. Computational corpus analysis shows that three institutional environments with high conceptual rigor — academic political philosophy, Catholic social doctrine, and the Inter-American Court of Human Rights — had independently exited the term "equity" before the DEI meme complex entered the lexical space (Lerer, 2026f). Political philosophy abandoned "equity" because the concept lacked the precision required by Rawlsian distributive justice analysis. The Vatican corpus shows zero uses of "equity" after 1965, operating instead on "justice" and "common good."



The IACtHR landmark gender-rights cases use equity zero times, operating on equality and non-discrimination.

The DEI meme colonized a term that multiple intellectual traditions had independently abandoned as insufficiently precise. Crucially, the abandonment by high-rigor environments was itself the condition of possibility for the colonization: if political philosophers were still actively disputing the meaning of equity as a technical term, the DEI meme would face competitive resistance from a well-organized semantic incumbent.

### **5.3 Why Imprecision Is Selected For**

Betzig's arbiters extracted maximum perquisites because they occupied the enforcement node. The DEI meme complex achieves high transmission fitness through a different mechanism: semantic imprecision maximizes the meme's fitness in high-competition, low-attention-span environments. Haidt's (2012) moral psychology identifies the mechanism: moral intuitions drive conclusions and reasoning follows. A meme that activates moral intuitions without specifying precise implications generates coalitional solidarity at lower cognitive cost than a meme requiring technical engagement with distributive justice theory.

Betzig's finding that incest taboos were enforced with the greatest severity when they protected the reproductive interests of the most powerful actors has an analogue here. The DEI meme complex is enforced with the greatest severity when violation threatens the coalitional cohesion of the implementing institution. The mechanism is not reproductive fitness but coalitional fitness: departure from the meme signals willingness to defect from the coalition, which imposes costs on the defector through the Lateral Silencing Exaptation (LSE) mechanism documented in EPT (Lerer, 2026b).

### **5.4 The LSE as the Modern Enforcement Mechanism**

Betzig documents that the most severe normative enforcement in pre-industrial polities targeted threats to the chief's reproductive monopoly: adultery with royal wives was capital in Ashanti, Bemba, Aztec, and Byzantine contexts simultaneously (Betzig, 1986, chapters 3–5). The enforcement severity was not proportional to the harm to third parties; it was proportional to the threat to the dominant actor's fitness.

The LSE mechanism operates analogously in contemporary regulatory and academic environments. Structurally unenforceable norms acquire a secondary function: they suppress political contestation by regulated agents who are permanently in technical violation (Lerer, 2026b). In the identity-politics context, the parallel is precise: agents who contest DEI frameworks do not necessarily violate any formal rule, but contestation exposes them to moral reclassification that imposes reputational costs through social network dynamics. The cost is not imprisonment; it is exclusion from the coalition.

The Betzig-EPT synthesis predicts that the enforcement severity of identity-norm violations will be disproportionate to any measurable harm to third parties and proportionate to the threat the violation poses to coalitional cohesion. This is falsifiable: we should expect, and the existing literature on cancellation events documents, that the harshest social sanctions attach to violations by high-status insiders rather than low-status outsiders, because high-status insider defection poses the greatest threat to coalition integrity.

## **6. Falsifiable Predictions**

The integration of Betzig's framework with EPT generates three predictions that extend prior EPT work and that are falsifiable on available or collectable evidence.

*Prediction 1.* In contemporary regulatory environments where enforcement agents have extractable perquisites, compliance documentation quality (volume, certification rates, formal program completeness) will be negatively correlated with actual behavioral compliance, not positively correlated. The mechanism is the same as Betzig's arbiters: actors controlling enforcement extract maximum perquisites while maintaining formal normative signals. This is distinguishable from the null hypothesis (no correlation) and from the enforcement-theory prediction (positive correlation). Relevant data: FCPA enforcement outcomes versus compliance program scores; Ley 27.401 certification rates versus enforcement actions.

*Prediction 2.* Lexical exit from high-precision normative concepts will precede rather than follow their colonization by low-precision meme complexes in high-competition environments. The Betzig mechanism — dominant actors occupy niches vacated by weaker ones — predicts that the exit occurs first, creating the available niche. The lexical exaptation data for "equity" support this prediction; replication on other contested normative concepts ("justice," "rights," "democracy") would provide additional tests.

*Prediction 3.* The severity of social sanctions for identity-norm violations will be predicted more accurately by the violator's institutional position (insider/outsider, status level) than by the content or demonstrable harm of the violation. Betzig's adultery-norm data show exactly this: the severity of punishment correlated with the chief's reproductive investment in the woman, not with community harm. The modern analogue predicts that cancellation events should show higher sanction rates for tenured professors than for unaffiliated commentators making identical statements, controlling for statement content.

## **7. Theoretical Limitations and Open Questions**

Three limitations of the present synthesis deserve explicit marking.

First, the Betzig-EPT bridge proposed here relies on structural analogy between genetic fitness dynamics and memetic fitness dynamics. The analogy is defended by the Generalized Darwinism program (Hodgson and Knudsen, 2010) and is foundational to EPT. But critics of memetic approaches (Sperber, 1996; Atran, 2001) argue that cultural transmission is too noisy, context-dependent, and reconstructive to support the replicator metaphor. The present paper does not resolve this debate; it operates within EPT's theoretical commitments while acknowledging that the meme-gene analogy is a working hypothesis, not an established equivalence.

Second, the LSE mechanism applied to identity politics generates predictions about enforcement patterns that are plausible and directionally supported by existing case evidence, but the systematic data required for formal hypothesis testing have not been assembled in a form that controls for the relevant confounds. The prediction in Section 6 specifies what that evidence would need to look like.

Third, the paper treats the welfare-state transition as a morphospace mutation, but the boundaries of the "gendarme" and "welfare" configurations in real institutional systems are not crisp. Most contemporary legal systems contain elements of both configurations, and the PSO dynamics Betzig documents appear in both. A more precise formalization would require a continuous measure of the constructivist-design intensity of specific institutional domains rather than a binary classification at the system level. The CLI framework developed in prior EPT work (Lerer, 2026d) may provide the instrument, but its application to this question has not been developed.

## **8. Conclusion**

Betzig's 1986 study has been cited in evolutionary psychology, behavioral ecology, and political anthropology. It has not been integrated with evolutionary theories of legal institutions, and the reasons for this are worth noting: the integration requires accepting that proto-legal systems encoded reproductive advantage for dominant actors as a feature rather than a malfunction, which sits uncomfortably with normative theories of law as coordination mechanism or social contract.

EPT's theoretical structure is not threatened by this finding; it is strengthened by it. The core EPT claim is that legal institutions are extended phenotypes of competing replicators subject to selection, not products of rational design serving general welfare. Betzig provides direct empirical evidence that this was true of proto-legal institutions in pre-industrial societies across six continents and multiple millennia. The gendarme state was a PSO from the beginning. The welfare state attempted to correct the extraction asymmetry and produced, predictably, a new configuration of the same underlying PSO dynamic operating through compliance theater rather than direct polygyny.

Twenty-first century identity claims represent neither a departure from nor a resolution of this dynamic. They are the latest instantiation of high-fitness coalitional signaling in an environment that has shifted the relevant fitness currency from reproductive access to institutional position. The mechanism Betzig identified in the courts of Ashanti and Byzantium — enforcement severity disproportionate to harm and proportionate to threat to dominant-actor fitness — is visible in the contemporary management of identity-norm violations by academic and corporate institutions. The form has changed. The selection dynamic has not.

Whether that dynamic is alterable — and under what conditions — is the question this synthesis leaves open, and deliberately so.

Betzig's own data contain a partial answer: despotic reproductive monopolies were historically constrained not when moral sensibilities improved but when subordinate coalition capacity exceeded dominant enforcement capacity, a structural threshold rather than a normative one.

EPT's equilibrium analysis suggests an analogue for the contemporary case: the fitness advantage of imprecision-based coalitional signaling erodes when institutional environments that reward conceptual precision compete effectively for the same actors — through jurisdictional competition, adversarial epistemic norms, or exit rights that make coalition defection less costly. The CLI framework developed in this research program provides one instrument for locating where on that arc a given institutional domain currently sits; a CLI above the ESS threshold of 0.71 signals lock-in, not irreversibility. These are, at present, hypotheses rather than findings. The conditions under which the selection dynamic itself comes under selection pressure remain an open empirical question for the program. Identifying it as such is not a concession to indeterminacy. It is the research agenda.

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## **LAKATOSIAN EPISTEMOLOGICAL POSITIONING DECLARATION**

This paper is situated within a Lakatosian progressive research program (Lakatos, 1978). The hard core consists of the claims that legal institutions are extended phenotypes of cultural replicators, that their persistence is explained by memetic fitness under selection pressure, and that the relevant fitness currency varies across institutional environments. The protective belt includes the specific mechanisms formalized here (PSO, LSE, Dennett-Nash Gap, Capacity-Gap) and the empirical indices derived from them (CLI, TCI). Following Popper (1959), the predictions in Section 6 are stated in falsifiable form; refutation of these predictions would require modification of the protective belt but not abandonment of the hard core unless the refutation pattern is systematic across multiple predictions simultaneously. Following Hull (1988) and Campbell (1974), the paper treats cultural selection as operating by mechanisms structurally analogous to biological selection without claiming they are identical.

## **AI DISCLOSURE**

Artificial intelligence tools were used for research support in this paper, including locating sources, extracting information from the EPT corpus notebooks, and reviewing drafts for stylistic consistency. All analysis, argumentation, theoretical synthesis, and conclusions are the author's own original contribution.

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